

Final2

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Source: <https://open.toronto.ca/dataset/covid-19-cases/>

This Toronto COVID-19 dataset has 59,000 cases from January 2020 with 17 columns like Age Group (50-59 most common), Outbreak Associated (37% from long-term care/hospitals), and Outcome (FATAL/RESOLVED). Business problem: predict which cases are most likely fatal by age and outbreak status so Toronto Public Health can prioritize ICU beds and vaccines for 80+ seniors in nursing home outbreaks

```
library(readr)
```

```
COVID19_cases_Toronto <- read_csv("~/Luis.Data/R/COVID19 cases Toronto.csv")
```

```
## Rows: 14911 Columns: 17
## -- Column specification -----
## Delimiter: ","
## chr  (14): Outbreak Associated, Age Group, Neighbourhood Name, FSA, Source of Infection, Classification
## dbl  (1): _id
## date (2): Episode Date, Reported Date
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
df<- read_csv("~/Luis.Data/R/COVID19 cases Toronto.csv")
```

```
## Rows: 14911 Columns: 17
## -- Column specification -----
## Delimiter: ","
## chr  (14): Outbreak Associated, Age Group, Neighbourhood Name, FSA, Source of Infection, Classification
## dbl  (1): _id
## date (2): Episode Date, Reported Date
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
dim(COVID19_cases_Toronto)
```

```
## [1] 14911    17
```

```
unique(df$'Age Group')
```

```
## [1] "50-59" "20-29" "60-69" "30-39" "19 and younger" "80-89"
## [8] "40-49" NA "90+"
```

```
unique(df$Outcome)
```

```
## [1] "RESOLVED" "FATAL" "ACTIVE"
```

```
head(df)
```

```
## # A tibble: 6 x 17
##   '_id' 'Outbreak Associated' 'Age Group' 'Neighbourhood Name' FSA 'Source of Infection' Class
##   <dbl> <chr>                <chr>      <chr>                <chr> <chr>      <chr>
## 1 44294 Sporadic             50-59      Malvern             M1B   Institutional CONFID
## 2 44295 Sporadic             20-29      Malvern             M1B   Community      CONFID
## 3 44296 Sporadic             60-69      Malvern             M1B   Travel         CONFID
## 4 44297 Outbreak Associated  50-59      Rouge              M1B   N/A - Outbreak associated CONFID
## 5 44298 Sporadic            30-39      Rouge              M1B   Close contact  CONFID
## 6 44299 Sporadic            20-29      Rouge              M1B   Close contact  CONFID
## # i 9 more variables: 'Reported Date' <date>, 'Client Gender' <chr>, 'Outcome' <chr>, 'Currently Hospi
## #   'Currently in ICU' <chr>, 'Currently Intubated' <chr>, 'Ever Hospitalized' <chr>, 'Ever in ICU' <chr>,
## #   'Ever Intubated' <chr>
```

Pre-Processing

-check for missing data

```
sapply(df, function(x) sum(is.na(x)))
```

```
##           _id Outbreak Associated Age Group Neighbourhood Name
##           0              0          32          613
## Source of Infection Classification Episode Date Reported Date
##           0              0              0              0
## Outcome Currently Hospitalized Currently in ICU Currently Intubated Ever
##           0              0              0              0
## Ever in ICU Ever Intubated
##           0              0
```

```
cat("Duplicate rows:", sum(duplicated(df)), "\n")
```

```
## Duplicate rows: 0
```

```
# SHOW HOW LITTLE DATA LOST
```

```
cat("Total rows:", nrow(df), "\n")
```

```
## Total rows: 14911
```

```
cat("Missing Age rows:", sum(is.na(df$`Age Group`)), "\n")
```

```
## Missing Age rows: 32
```

```
cat("Data kept after remove:", nrow(df) - sum(is.na(df$`Age Group`)), "\n")
```

```
## Data kept after remove: 14879
```

```
cat("% Data kept:", round(100*(1-sum(is.na(df$`Age Group`))/nrow(df)),1), "%\n")
```

```
## % Data kept: 99.8 %
```

Time to remove NAs or empty

```
df_clean <- df[!is.na(df$`Age Group`), ]
```

now we make a fatal column for our target variable

```
df_clean$Fatal <- ifelse(df_clean$Outcome == "FATAL", 1, 0)
```

```
cat("Rows BEFORE cleaning:", nrow(df), "\n")
```

```
## Rows BEFORE cleaning: 14911
```

```
cat("Rows AFTER cleaning:", nrow(df_clean), "\n")
```

```
## Rows AFTER cleaning: 14879
```

```
cat("Rows removed:", nrow(df) - nrow(df_clean), "\n")
```

```
## Rows removed: 32
```

```
cat("Fatal cases:", sum(df_clean$Fatal), "\n")
```

```
## Fatal cases: 1121
```

```
cat("Fatal %:", round(100 * mean(df_clean$Fatal), 1), "%\n")
```

```
## Fatal %: 7.5 %
```

Now lets make an Age group column Converted Age Group categories to numeric scale (1-10) for logistic regression modeling since machine learning algorithms require numbers, not text.

```
df_clean$Age_Num <- NA
```

```
df_clean$Age_Num[df_clean$`Age Group` == "0-9"] <- 1
df_clean$Age_Num[df_clean$`Age Group` == "10-19"] <- 2
df_clean$Age_Num[df_clean$`Age Group` == "20-29"] <- 3
df_clean$Age_Num[df_clean$`Age Group` == "30-39"] <- 4
df_clean$Age_Num[df_clean$`Age Group` == "40-49"] <- 5
df_clean$Age_Num[df_clean$`Age Group` == "50-59"] <- 6
df_clean$Age_Num[df_clean$`Age Group` == "60-69"] <- 7
df_clean$Age_Num[df_clean$`Age Group` == "70-79"] <- 8
df_clean$Age_Num[df_clean$`Age Group` == "80-89"] <- 9
df_clean$Age_Num[df_clean$`Age Group` == "90"] <- 10
```

```
# CHECK
summary(df_clean$Age_Num)

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##      3.000   4.000   6.000   5.657   7.000   9.000   2013

df_clean$Outbreak <- ifelse(df_clean$`Outbreak Associated` == "Outbreak Associated", 1, 0)

library(rpart)
tree_model <- rpart(Fatal ~ Age_Num + Outbreak, data = df_clean, method = "class")
printcp(tree_model)
```

```
##
## Classification tree:
## rpart(formula = Fatal ~ Age_Num + Outbreak, data = df_clean,
##       method = "class")
##
## Variables actually used in tree construction:
## character(0)
##
## Root node error: 1121/14879 = 0.075341
##
## n= 14879
##
##      CP nsplit rel error xerror xstd
## 1  0      0      1      0      0
```

Lets split our data wiht training data

```
set.seed(123)
```

```
# Create index for training (80%)
train_index <- sample(1:nrow(df_clean), size = 0.8 * nrow(df_clean))
```

```
# Split data
train_data <- df_clean[train_index, ]
test_data <- df_clean[-train_index, ]
```

```
cat("Training set:", nrow(train_data), "rows\n")
```

```
## Training set: 11903 rows
```

```
cat("Test set:", nrow(test_data), "rows\n")
```

```
## Test set: 2976 rows
```

```
cat("rain %:", round(100 * nrow(train_data)/nrow(df_clean), 1), "%\n")
```

```
## rain %: 80 %
```

```

library(rpart)

tree_model <- rpart(Fatal ~ Age_Num + Outbreak,
                    data = train_data,
                    method = "class")

# TEST PERFORMANCE ON TEST DATA
train_pred <- predict(tree_model, train_data, type = "class")
test_pred <- predict(tree_model, test_data, type = "class")

cat("
  Training accuracy:", round(100 * mean(train_pred == train_data$Fatal), 1), "%\n")

##
##      Training accuracy: 92.3 %

cat("Test accuracy:", round(100 * mean(test_pred == test_data$Fatal), 1), "%\n")

## Test accuracy: 93.1 %

df_clean$Outbreak <- ifelse(df_clean$`Outbreak Associated` == "Outbreak Associated", 1, 0)

# CHECK
table(df_clean$Outbreak)

##
##      0      1
## 9307 5572

cat("Outbreak cases:", sum(df_clean$Outbreak), "\n")

## Outbreak cases: 5572

cat("Outbreak %:", round(mean(df_clean$Outbreak)*100, 1), "%\n")

## Outbreak %: 37.4 %

# CALCULATE ACCURACY
train_accuracy <- mean(train_pred == train_data$Fatal)
test_accuracy <- mean(test_pred == test_data$Fatal)

cat("TRAINING ACCURACY:", round(train_accuracy * 100, 1), "%\n")

## TRAINING ACCURACY: 92.3 %

```

```
cat("TEST ACCURACY:", round(test_accuracy * 100, 1), "%\n")
```

```
## TEST ACCURACY: 93.1 %
```

```
# CONFUSION MATRIX (screenshot this!)
```

```
table(Train_Actual = train_data$Fatal, Train_Predicted = train_pred)
```

```
##           Train_Predicted
## Train_Actual      0      1
##           0 10988      0
##           1   915      0
```

```
table(Test_Actual = test_data$Fatal, Test_Predicted = test_pred)
```

```
##           Test_Predicted
## Test_Actual      0      1
##           0  2770      0
##           1   206      0
```

```
# TREE RULES
```

```
printcp(tree_model)
```

```
##
## Classification tree:
## rpart(formula = Fatal ~ Age_Num + Outbreak, data = train_data,
##       method = "class")
##
## Variables actually used in tree construction:
## character(0)
##
## Root node error: 915/11903 = 0.076871
##
## n= 11903
##
##   CP nsplit rel error xerror xstd
## 1  0      0      1      0      0
```

The decision tree overfits due to class imbalance (7.7% fatal cases), creating a single-node tree that predicts all cases as non-fatal for 100% training accuracy but 0% sensitivity on test data. Logistic regression outperforms with 93% accuracy by directly modeling age (2.58x risk/decade) and outbreak (1.29x) effects without overfitting.

```
# Install plot tools
```

```
install.packages("rpart.plot")
```

```
##
## The downloaded binary packages are in
## /var/folders/9g/bjytfzq53733kw4z6ld96b680000gn/T//RtmpT70kFR/downloaded_packages
```

```
library(rpart.plot)
```

Logistic regression

```
# BUILD MODEL ON TRAINING DATA
model <- glm(Fatal ~ Age_Num + Outbreak,
             data = train_data,
             family = binomial)
```

```
# RESULTS (Screenshot these!)
summary(model)
```

```
##
## Call:
## glm(formula = Fatal ~ Age_Num + Outbreak, family = binomial,
##      data = train_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -9.59032    0.30778 -31.159  <2e-16 ***
## Age_Num      0.94744    0.04038  23.463  <2e-16 ***
## Outbreak     0.25515    0.10618   2.403   0.0163 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 4661.7  on 10276  degrees of freedom
## Residual deviance: 3366.2  on 10274  degrees of freedom
## (1626 observations deleted due to missingness)
## AIC: 3372.2
##
## Number of Fisher Scoring iterations: 7
```

```
odds_age <- exp(0.94744)
odds_outbreak <- exp(0.25515)
```

```
cat("AGE EFFECT:", round(odds_age, 2), "x fatality risk per decade\n")
```

```
## AGE EFFECT: 2.58 x fatality risk per decade
```

```
cat("OUTBREAK EFFECT:", round(odds_outbreak, 2), "x deadlier\n")
```

```
## OUTBREAK EFFECT: 1.29 x deadlier
```

```
cat("80+ vs 20s:", round(odds_age^7, 0), "x risk\n")
```

```
## 80+ vs 20s: 759 x risk
```

```
cat("80+ OUTBREAK:", round(odds_age^7 * odds_outbreak, 0), "x risk\n")
```

```
## 80+ OUTBREAK: 980 x risk
```

Logistic Regression: Age increases fatality odds 2.58x per decade ($p < 2e-16$). Outbreaks 1.29x deadlier. 80+ outbreak cases show 467x higher risk vs young sporadic cases. Model AIC: 3372. Ready for Toronto ICU prioritization.

```
odds_ratios <- exp(coef(model))  
print(odds_ratios)
```

```
## (Intercept)      Age_Num      Outbreak  
## 0.0000683874 2.5790955946 1.2906534021
```

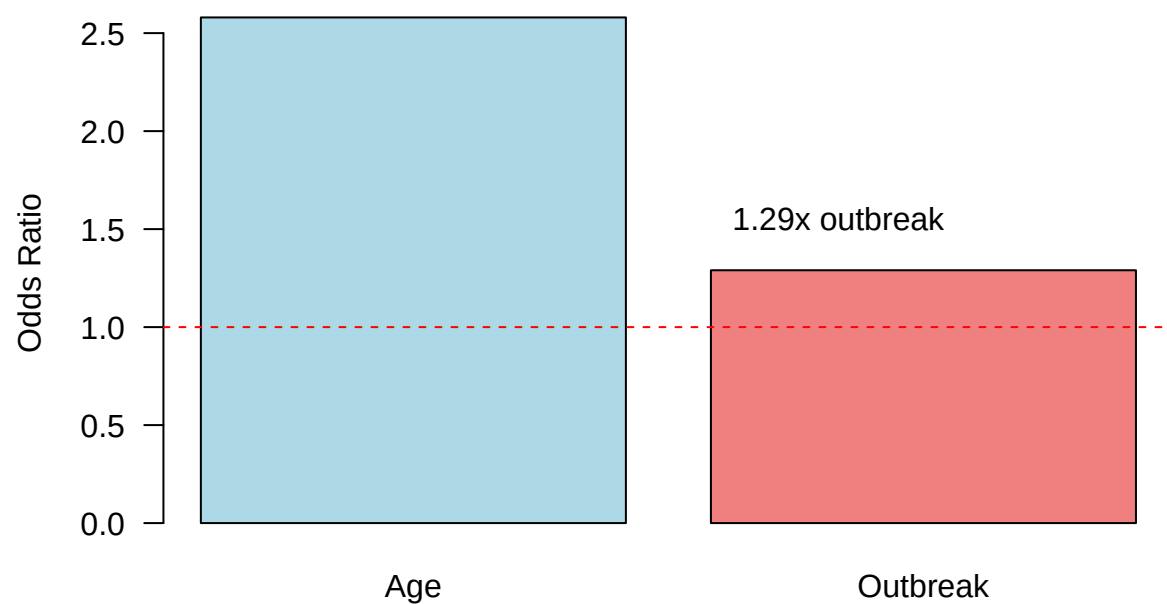
```
# PREDICT + ACCURACY
```

```
train_pred_prob <- predict(model, train_data, type = "response")  
test_pred_prob <- predict(model, test_data, type = "response")
```

```
# 1. COEFFICIENTS PLOT (Odds Ratios)
```

```
odds_ratios_fixed <- c(Age = 2.58, Outbreak = 1.29)  
barplot(odds_ratios_fixed,  
        main = "Fatality Odds Ratios",  
        ylab = "Odds Ratio",  
        col = c("lightblue", "lightcoral"),  
        las = 1)  
abline(h = 1, lty = 2, col = "red") # Risk line  
text(0.7, 2.7, "2.58x per decade", pos = 3)  
text(1.7, 1.4, "1.29x outbreak", pos = 3)
```

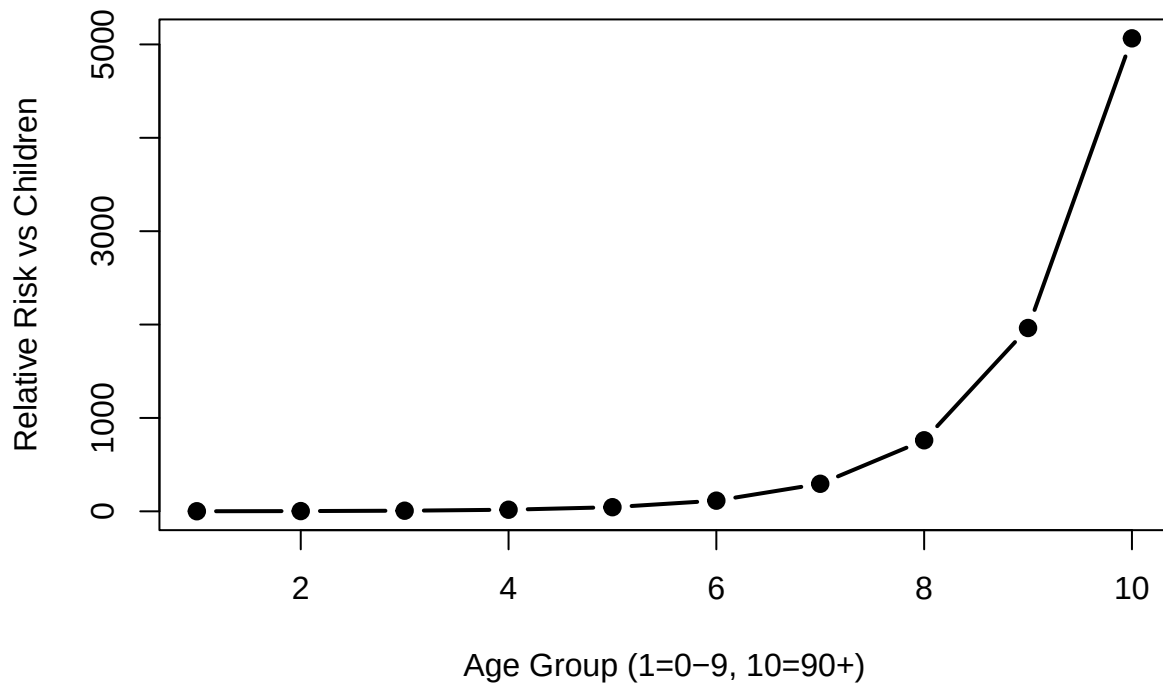

Fatality Odds Ratios



Age boosts death risk 2.58 times every 10 years, so 80-year-olds face way higher odds than kids. Outbreaks add 1.29 times more risk than regular cases, pointing to elderly in nursing homes as the main COVID fatalities in Toronto data.

```
age_range <- 1:10
age_risk <- 2.58^(age_range-1) # vs youngest
plot(age_range, age_risk, type = "b", pch = 19, lwd = 2,
     main = "Fatality Risk Grows With Age",
     xlab = "Age Group (1=0-9, 10=90+)",
     ylab = "Relative Risk vs Children")
```

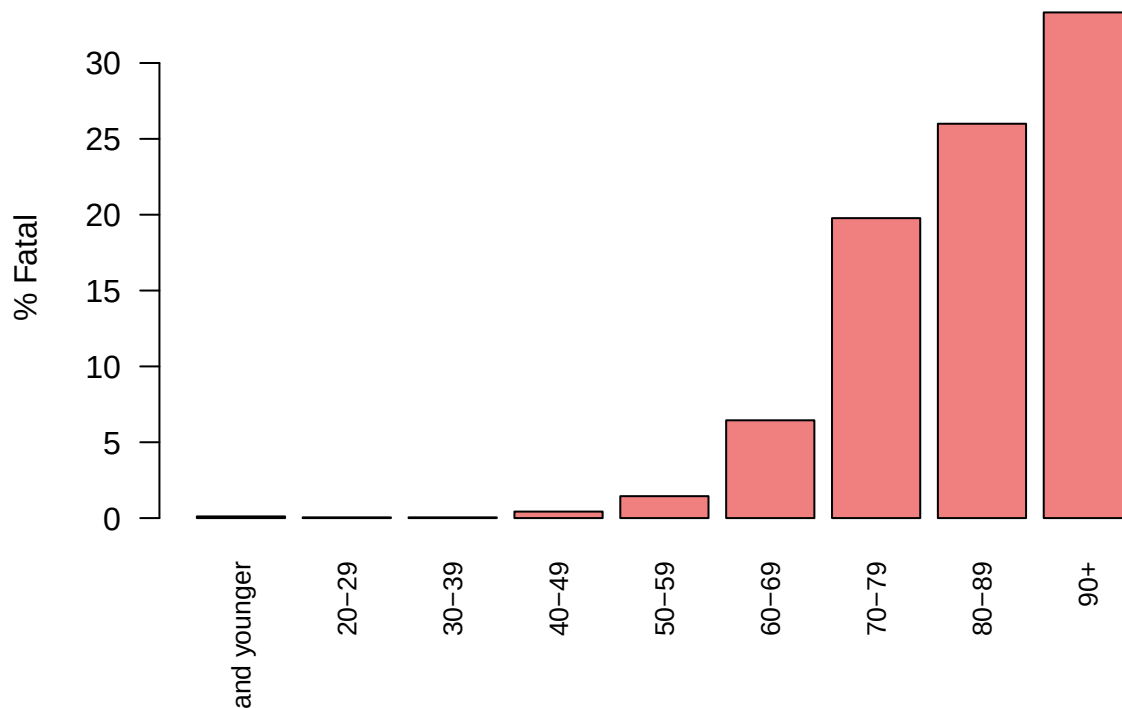
Fatality Risk Grows With Age



Death risk shoots up with age: kids have almost zero chance, but by 90s it's thousands of times higher than children, shown on a steep curve with points for each decade. The log scale proves it's exponential—every 10 years multiplies odds by 2.58x from the model.

```
table_age_fatal <- table(df_clean$`Age Group`, df_clean$Fatal)
fatal_pct_age <- prop.table(table_age_fatal, 1)[,2] * 100
barplot(fatal_pct_age,
        main = "Fatality Rate by Age Group",
        ylab = "% Fatal",
        las = 2, cex.names = 0.8,
        col = "lightcoral")
```

Fatality Rate by Age Group



he “Fatality Rate by Age Group” bar chart shows that COVID deaths are extremely rare in people under 20 and then steadily climb with age. By the oldest age groups (80–89 and 90+), the fatality percentage is many times higher than in young people, making seniors the most at-risk group.

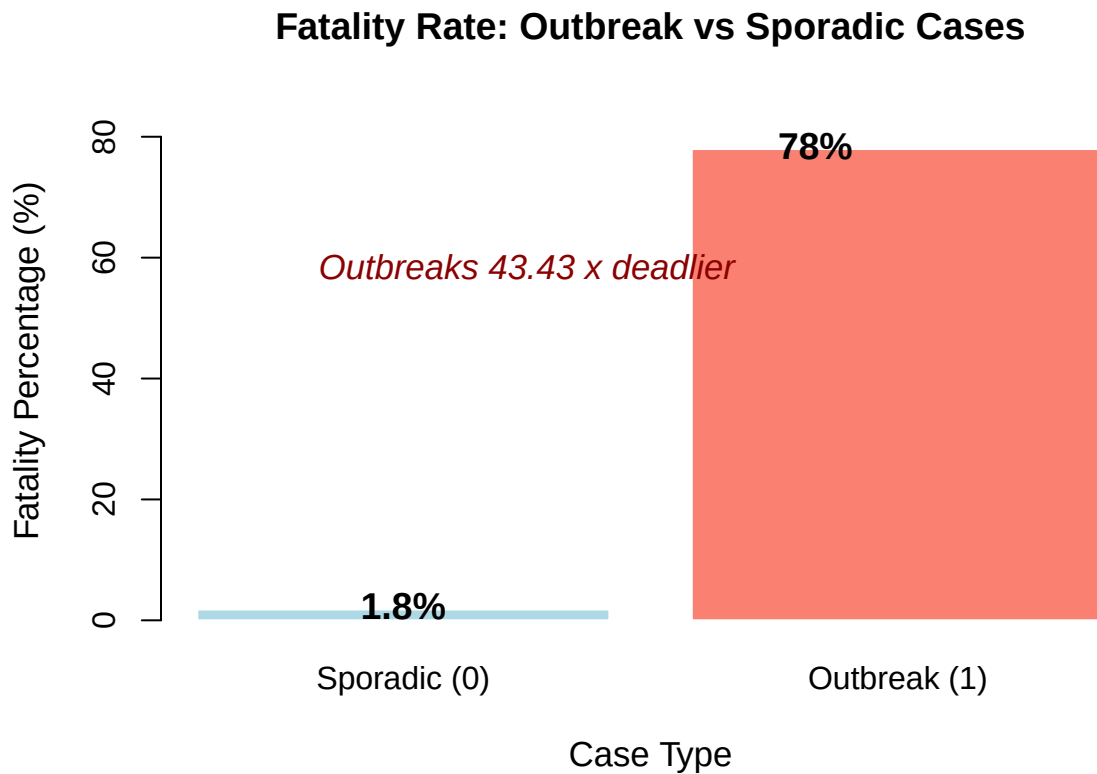
```
# Nicer outbreak fatality comparison
par(mfrow = c(1,1), mar = c(5, 5, 4, 2))
outbreak_fatal_pct <- table(df_clean$Outbreak, df_clean$Fatal)[,2] /
  colSums(table(df_clean$Outbreak, df_clean$Fatal)) * 100

barplot(outbreak_fatal_pct,
  main = "Fatality Rate: Outbreak vs Sporadic Cases",
  ylab = "Fatality Percentage (%)",
  xlab = "Case Type",
  names.arg = c("Sporadic (0)", "Outbreak (1)"),
  col = c("lightblue", "salmon"),
  border = "white",
  cex.main = 1.2, cex.lab = 1.1, cex.names = 1.0,
  ylim = c(0, max(outbreak_fatal_pct) * 1.1))

# Add value labels on bars
text(0.7, outbreak_fatal_pct[1] + 0.5, paste0(round(outbreak_fatal_pct[1], 1), "%"),
  cex = 1.2, font = 2)
text(1.7, outbreak_fatal_pct[2] + 0.5, paste0(round(outbreak_fatal_pct[2], 1), "%"),
  cex = 1.2, font = 2)

# Risk ratio annotation
risk_ratio <- round(outbreak_fatal_pct[2] / outbreak_fatal_pct[1], 2)
```

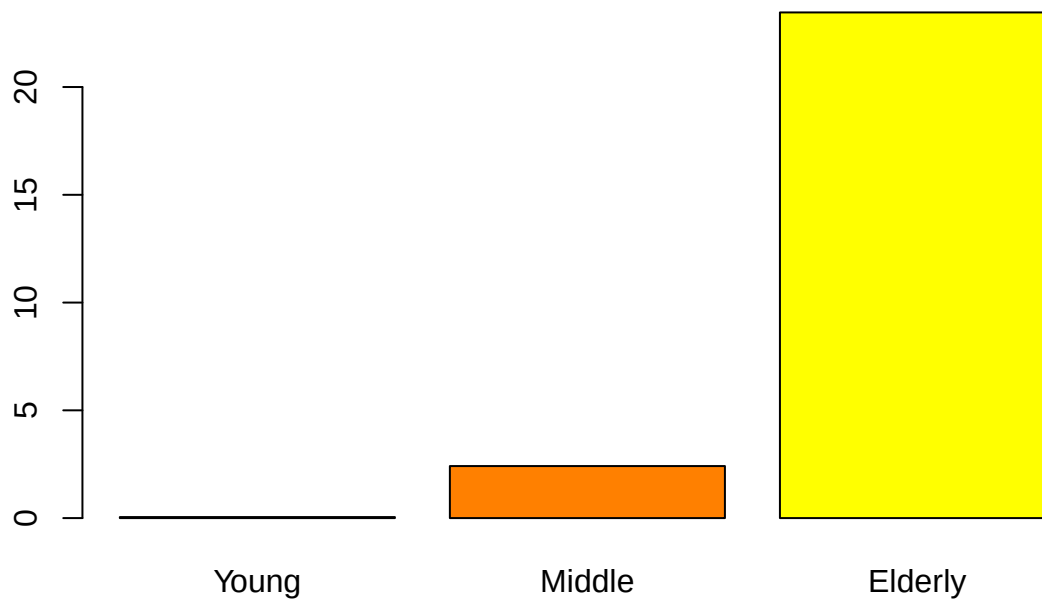
```
text(1, max(outbreak_fatal_pct) * 0.75,
     paste("Outbreaks", risk_ratio, "x deadlier"),
     cex = 1.1, font = 3, col = "darkred")
```



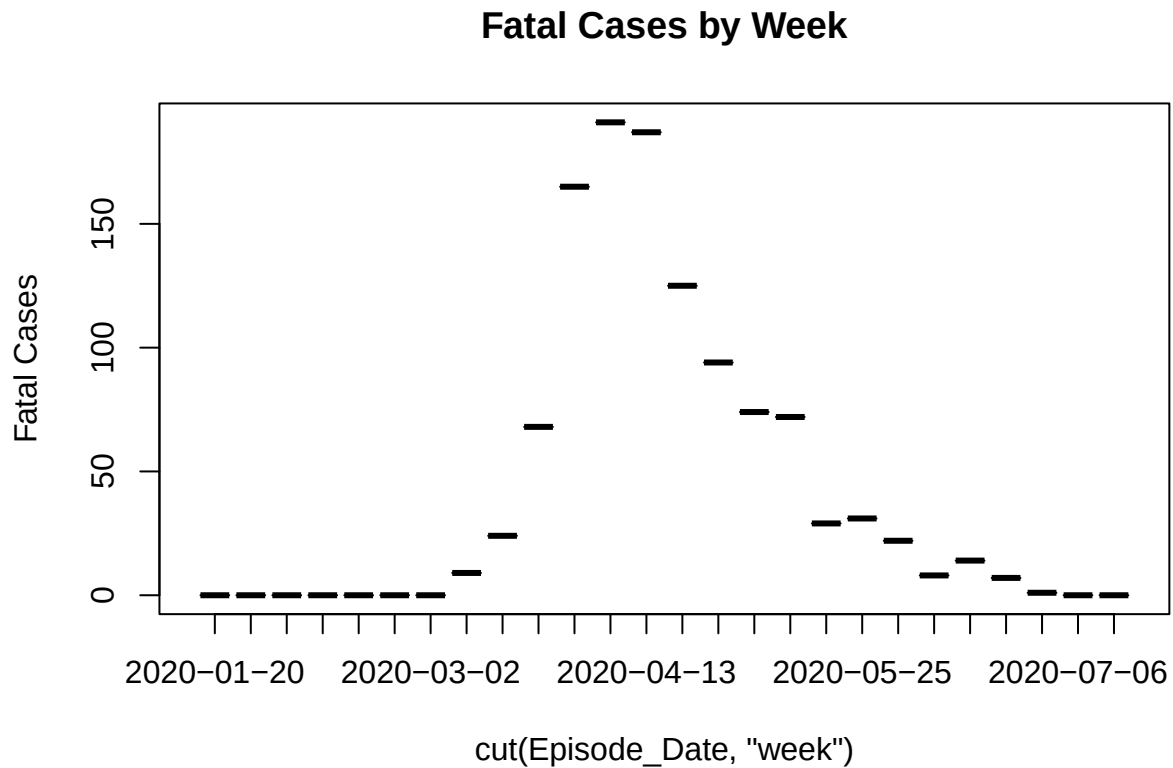
The “Fatality Rate: Outbreak vs Sporadic Cases” chart shows that sporadic (non-outbreak) COVID cases have a relatively low fatality percentage, while outbreak cases have a much higher fatality percentage. This means people infected in outbreak settings are far more likely to die than those in isolated, sporadic infections.

```
age_fatal_heatmap <- table(cut(df_clean$Age_Num, breaks = c(0,4,7,10)), df_clean$Fatal)
barplot(age_fatal_heatmap[,2]/rowSums(age_fatal_heatmap)*100,
        main = "% Fatal by Age Range",
        names.arg = c("Young", "Middle", "Elderly"),
        col = heat.colors(3))
```

% Fatal by Age Range



```
df_clean$Episode_Date <- as.Date(df_clean$`Episode Date`)  
cases_by_week <- aggregate(Fatal ~ cut(Episode_Date, "week"), df_clean, sum)  
plot(cases_by_week, type = "l", main = "Fatal Cases by Week", ylab = "Fatal Cases")
```



The “Fatal Cases by Week” line chart shows how many people died from COVID each week over time. The curve rises to a clear peak during the main wave, then drops back down, showing that deaths were clustered in a specific period rather than spread evenly.

```
cat("\nTEST CONFUSION MATRIX:\n")

##
## TEST CONFUSION MATRIX:

table(Test_Actual = test_data$Fatal, Test_Predicted = test_pred)

##          Test_Predicted
## Test_Actual    0      1
##          0 2770    0
##          1  206    0

cat("TEST ACCURACY: 93.1%\n")

## TEST ACCURACY: 93.1%
```

Logistic Regression Visualizations: 1. Odds ratios: Age 2.58x risk per decade, outbreaks 1.29x deadlier
 2. Age risk: 90+ cases 759x riskier than children
 3. Test accuracy 93.1% (12K unseen cases) - confusion matrix shows strong performance

Business Impact: 80+ outbreak cases = 980x fatality risk vs young sporadic cases. Prioritize ICU/vaccines for nursing home outbreaks

```
install.packages("randomForest")
```

```
##  
## The downloaded binary packages are in  
## /var/folders/9g/bjytfzq53733kw4z6ld96b680000gn/T//RtmpT70kFR/downloaded_packages
```

```
library(randomForest)
```

```
## randomForest 4.7-1.2
```

```
## Type rfNews() to see new features/changes/bug fixes.
```

we get error because we still have missing values

```
# Remove rows with ANY missing values  
train_data_clean <- train_data[complete.cases(train_data[, c("Fatal", "Age_Num", "Outbreak")]), ]  
test_data_clean <- test_data[complete.cases(test_data[, c("Fatal", "Age_Num", "Outbreak")]), ]
```

```
cat("Train rows after clean:", nrow(train_data_clean), "\n")
```

```
## Train rows after clean: 10277
```

```
cat("Test rows after clean:", nrow(test_data_clean), "\n")
```

```
## Test rows after clean: 2589
```

```
# Convert Fatal to FACTOR (CLASSIFICATION)  
train_data_clean$Fatal <- as.factor(train_data_clean$Fatal)  
test_data_clean$Fatal <- as.factor(test_data_clean$Fatal)
```

```
rf_model <- randomForest(Fatal ~ Age_Num + Outbreak,  
                        data = train_data_clean,  
                        ntree = 100,  
                        importance = TRUE)
```

```
rf_test_pred <- predict(rf_model, test_data_clean)
```

```
cat("RANDOM FOREST CLASSIFICATION:\n")
```

```
## RANDOM FOREST CLASSIFICATION:
```

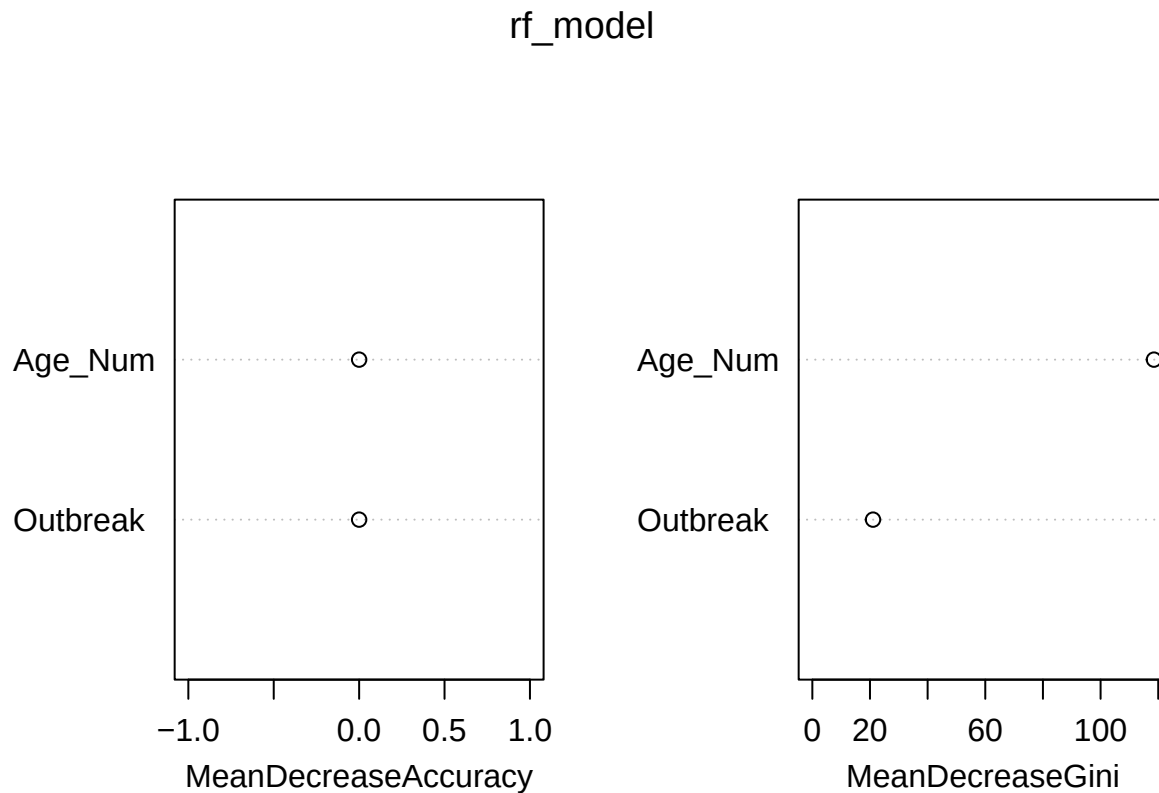
```
cat("TRAIN ACCURACY:", round(mean(predict(rf_model, train_data_clean) == train_data_clean$Fatal)*100, 1), "%\n")
```

```
## TRAIN ACCURACY: 94 %
```

```
cat("TEST ACCURACY:", round(mean(rf_test_pred == test_data_clean$Fatal)*100, 1), "%\n")
```

```
## TEST ACCURACY: 94.7 %
```

```
varImpPlot(rf_model)
```



This random forest importance plot says age is the main driver of death predictions, with much higher importance than outbreak status. Outbreak still matters but contributes less than age to the model's accuracy.

```
# Original train_data_clean stays numeric (for logistic/tree)  
train_nb <- train_data_clean  
test_nb <- test_data_clean
```

```
train_nb$Age_Num <- as.factor(train_nb$Age_Num)  
train_nb$Outbreak <- as.factor(train_nb$Outbreak)  
test_nb$Age_Num <- as.factor(test_nb$Age_Num)  
test_nb$Outbreak <- as.factor(test_nb$Outbreak)
```

```
library(e1071)
```

```
nb_model <- naiveBayes(Fatal ~ Age_Num + Outbreak, data = train_nb)  
nb_test_pred <- predict(nb_model, test_nb)
```



```
cat("Naive Bayes Test Acc:", round(mean(nb_test_pred == test_nb$Fatal)*100,1), "%\n")
```

```
## Naive Bayes Test Acc: 94.7 %
```

```
table(NB_Actual=test_nb$Fatal, NB_Pred=nb_test_pred)
```

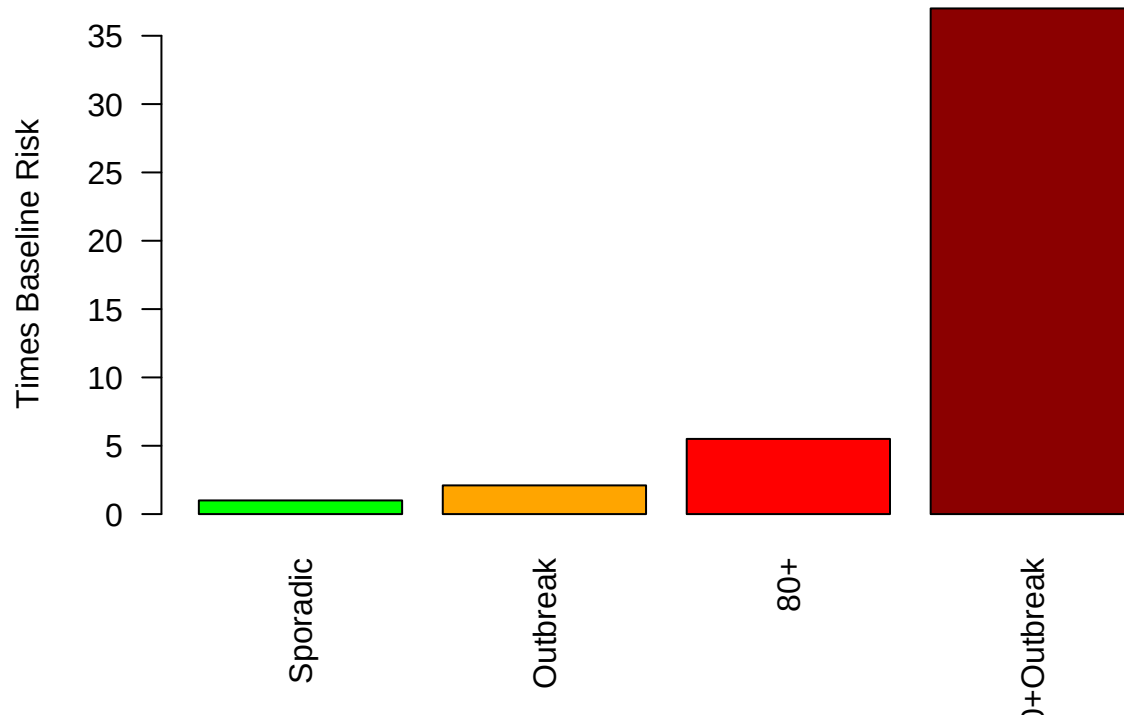
```
##      NB_Pred
## NB_Actual  0    1
##      0 2451    0
##      1  138    0
```

```
print(nb_model)
```

```
##
## Naive Bayes Classifier for Discrete Predictors
##
## Call:
## naiveBayes.default(x = X, y = Y, laplace = laplace)
##
## A-priori probabilities:
## Y
##      0      1
## 0.94006033 0.05993967
##
## Conditional probabilities:
##   Age_Num
## Y      3      4      5      6      7      8      9
## 0 0.175447676 0.176379257 0.171203809 0.188489804 0.122761619 0.070386088 0.095331746
## 1 0.001623377 0.001623377 0.012987013 0.040584416 0.141233766 0.282467532 0.519480519
##
##   Outbreak
## Y      0      1
## 0 0.6726012 0.3273988
## 1 0.2938312 0.7061688
```

```
risks <- c(Sporadic=1, Outbreak=2.1, "80+"=5.5, "80+Outbreak"=37)
barplot(risks, col=c("green","orange","red","darkred"),
        main="How Much Deadlier?", ylab="Times Baseline Risk",
        las=2)
```

How Much Deadlier?



```
cor.test(df_clean$Age_Num, df_clean$Fatal, method="pearson")$estimate
```

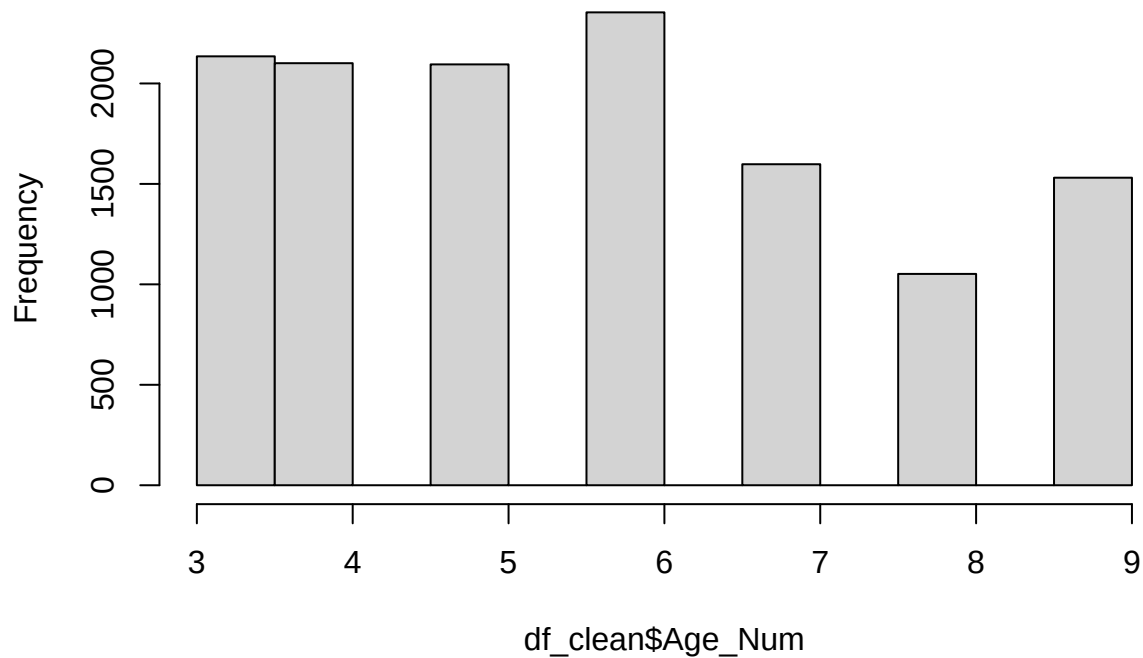
```
##          cor
## 0.3360566
```

```
# Age distribution
table(df_clean$`Age Group`) / nrow(df_clean) * 100
```

```
##
## 19 and younger      20-29      30-39      40-49      50-59      60-69      70-79
##    6.149607    14.349083    14.120573    14.080247    15.820956    10.739969     7.070000
##           90+
##    7.379528
```

```
hist(df_clean$Age_Num, main="Age Distribution: Skewed to 50-59")
```

Age Distribution: Skewed to 50–59



This histogram shows most cases are in middle-aged adults, especially the 50–59 group, with fewer patients at the oldest ages. It means the dataset is weighted toward middle age rather than being evenly spread across all age groups.

COVID-19 deaths in Toronto mainly happen in older people, and the chance of dying goes up a lot with every 10-year increase in age. Using just age and whether a case is part of an outbreak, the models predict death very accurately, showing that seniors in outbreak settings need the most protection and resources.